

A Three-Angle Framework for Spatial-Context-Augmented Causal Inference in Geographic Analysis

Ning Zhou^a and Xiang Jing^{a**}

^a*School of Software and Microelectronics, Peking University, Beijing 100871, China*

Abstract

Causal inference in geographic information science is often limited by spatial confounding, observational treatment assignment, and mechanisms that operate across multiple spatial scales. Statistical estimators can quantify treatment effects, but they often depend on incomplete covariate adjustment; knowledge-based reasoning can make causal mechanisms explicit, but it does not by itself identify effect sizes; and spatial simulation can explore interventions, but it requires empirical calibration.

We propose a three-angle framework for spatial-context-augmented causal analysis. *Angle A* provides statistical estimators, including propensity score matching, exposure-response functions, difference-in-differences, spatial Granger causality, geographic convergent cross-mapping, and causal forests. These estimators can be augmented with spatial-context representations, including geospatial foundation model embeddings when available and remote-sensing-derived covariates when foundation embeddings are not available for the study region. *Angle B* uses structured LLM-assisted reasoning to generate causal DAGs, counterfactual chains, mechanism explanations, and scenario descriptions. *Angle C* links these scenarios to geospatial world-model simulations and calibrates scenario magnitudes using empirical treatment-effect estimates from Angle A.

Controlled synthetic scenarios and a Chongqing urban heat island case study illustrate how the framework can expose spatial confounding and connect effect estimation with mechanism interpretation. The Chongqing case shows that a naive positive association between high-rise buildings and land surface temperature can reverse after spatial-context adjustment, but the current case uses Sentinel-2 and DEM features rather than AlphaEarth embeddings. We therefore treat the case as evidence for the broader spatial-context augmentation strategy, not as definitive validation of GeoFM embeddings. Additional multi-seed benchmarking, real GeoFM ablations, and spatial robustness tests are required before the framework’s stronger GeoFM-specific claims can be established.

^{**}Corresponding author. Email: jingxiang@pku.edu.cn

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1 Introduction

Understanding cause-and-effect relationships in geographic systems is essential for evidence-based policy in urban planning, environmental management, and public health [Verburg et al., 2004]. Yet causal inference in geography faces distinctive challenges that limit the applicability of methods developed for randomized experiments or simple observational studies.

First, *spatial confounding* arises because geographic units are embedded in continuous environmental fields—climate, soil, elevation, land-use context—that simultaneously influence both treatment assignment and outcomes. Traditional approaches control for measured confounders via regression or matching, but unmeasured spatial factors (collectively termed “spatial confounders”) remain a persistent threat to identification [Pearl, 2009]. Second, *limited experimental control* means that geographic interventions (infrastructure projects, conservation policies, land-use regulations) are rarely randomly assigned; researchers must rely on quasi-experimental designs that require careful assumption validation. Third, *multi-scale mechanisms* connect local actions to regional outcomes through complex spatial interactions (spillover effects, diffusion, agglomeration) that resist simple parametric modeling.

The recent convergence of three technological developments creates an opportunity to address these challenges holistically. Geospatial foundation models (GeoFMs) such as AlphaEarth [Brown et al., 2025] provide dense spatial representations that may encode land cover, terrain, climate, spectral properties, and other environmental context. Where such embeddings are available and valid for the study region, they can serve as learned proxies for spatial context; where they are unavailable, remote-sensing-derived covariates such as Sentinel-2 spectral indices and DEM features provide a more conventional fallback. Large language models (LLMs) such as Gemini 2.5 [Team Gemini, 2024] have demonstrated the ability to organize scientific mechanisms, construct causal graphs from domain knowledge, and generate structured counterfactual narratives. Geospatial world models [Zhou and Jing, 2026] can predict future land-use change in a compressed embedding space, enabling interventional simulations that project “what would happen if” a specific policy were applied to a sub-region.

Our key insight is that these three paradigms are *complementary*: each compensates for the others’ weaknesses. Statistical methods provide quantitative estimates but cannot explain *why* an effect occurs; LLM reasoning can articulate mechanisms and identify candidate confounders but cannot identify effect sizes on its own; world models can simulate interventions spatially but need empirical estimates to anchor their predictions to observed data. By bridging these three angles through shared spatial-context representations, a common data catalog, and explicit cross-angle interfaces, we create a system where each angle both

consumes and constrains the others’ outputs.

Our contributions are:

1. A three-angle framework for geographic causal inference that connects statistical, knowledge-based, and simulation-based approaches through shared spatial-context representations.
2. A spatial-context augmentation strategy in which GeoFM embeddings, when available, or remote-sensing-derived covariates can be used as learned or observed proxies for spatial confounding.
3. Cross-angle interfaces in which LLM-assisted scenarios map to world model inputs and empirical ATT estimates are used to calibrate scenario magnitudes.
4. An implementation with 14 tool functions and controlled validation scenarios that support reproducible testing, while identifying the additional real-world ablations needed for submission-grade evidence.

2 Related Work

2.1 Causal inference in geographic contexts

The application of causal inference methods to spatial data has a growing but fragmented literature. Propensity score matching (PSM) with spatial distance weighting has been applied to evaluate urban greening interventions [Stuart, 2010]. Spatial difference-in-differences (DiD) designs exploit policy discontinuities at administrative boundaries [Angrist and Pischke, 2009]. Granger causality has been extended to spatial panel data to detect temporal causal relationships between urbanization and environmental change [Granger, 1969]. Convergent cross-mapping (CCM), originally developed for ecological time series [Sugihara et al., 2012], has been adapted to spatial cross-sections through geographic weighting [Clark et al., 2015]. Causal forests [Athey and Imbens, 2019] enable heterogeneous treatment effect estimation across spatial units but do not inherently account for spatial dependence.

A common limitation across these methods is their treatment of spatial confounding: most rely on observed covariates (distance to city center, elevation, population density) as proxies, leaving unmeasured environmental factors uncontrolled. Our framework addresses this gap by making spatial-context representation an explicit part of the adjustment strategy. Depending on data availability, this context may be represented by GeoFM embeddings or by remote-sensing-derived covariates such as spectral indices, terrain variables, and land-cover descriptors.

2.2 LLMs for scientific causal reasoning

Recent work has demonstrated that LLMs can construct causal DAGs from natural language descriptions [Kiciman et al., 2023], generate counterfactual

explanations [Chen et al., 2024], and assist with variable selection in causal studies. However, LLM-generated causal structures lack the quantitative rigor needed for policy decisions: they can identify plausible confounders but cannot estimate effect magnitudes. Our Angle B addresses this limitation by explicitly bridging LLM outputs to statistical estimation (Angle A) and simulation (Angle C).

2.3 World models for interventional reasoning

World models—learned simulators that predict environment state transitions in compressed latent spaces—originated in reinforcement learning [Ha and Schmidhuber, 2018, Hafner et al., 2023]. Zhou and Jing [2026] introduced a geospatial world model using frozen AlphaEarth embeddings with a lightweight Latent-DynamicsNet for LULC change prediction. Our Angle C extends this work by adding interventional capabilities: sub-region scenario blending, spillover analysis, and treatment effect measurement in embedding space.

3 System Architecture

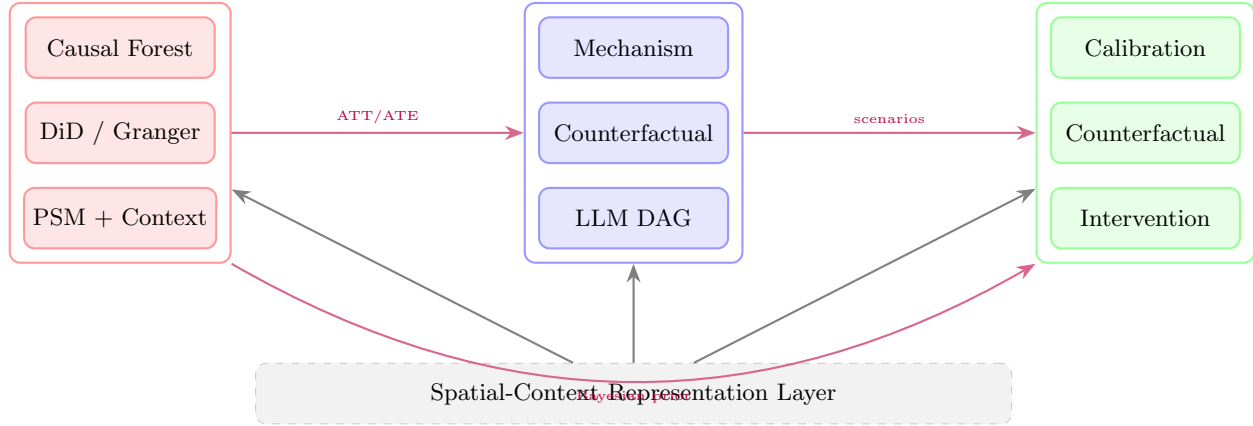


Figure 1: Three-angle causal inference framework. All three angles share a spatial-context representation layer, which can be instantiated with GeoFM embeddings or remote-sensing-derived covariates depending on data availability. Purple arrows indicate cross-angle bridges: Angle A provides ATT estimates to Angle B for interpretation and to Angle C for Bayesian calibration; Angle B generates structured scenarios consumable by Angle C.

The framework is implemented as three complementary toolsets within a GIS agent platform built on Google Agent Development Kit (ADK). Each angle is realized as a set of tool functions (Table 1) that can be invoked independently or composed through cross-angle bridges.

Table 1: Tool inventory across the three angles.

Angle	Tool	Key capability	Lines
A: Statistical	<code>propensity_score_matching</code>	ATE/ATT with GeoFM confounders	192
	<code>exposure_response_function</code>	Continuous dose-response via GPS	185
	<code>difference_in_differences</code>	Panel DiD with parallel trends test	137
	<code>spatial_granger_causality</code>	VAR-based temporal causal detection	151
	<code>geographic_causal_mapping</code>	GCCM convergence for nonlinear causality	225
	<code>causal_forest_analysis</code>	Heterogeneous CATE via T-learner	180
B: LLM	<code>construct_causal_dag</code>	Knowledge-driven DAG with 5 domains	88
	<code>counterfactual_reasoning</code>	Multi-step causal chain generation	82
	<code>explain_causal_mechanism</code>	Statistical result interpretation	74
	<code>generate_what_if_scenarios</code>	Structured scenario generation	61
C: World Model	<code>intervention_predict</code>	Sub-region intervention + spillover	219
	<code>counterfactual_comparison</code>	Dual-scenario parallel rollout	226
	<code>embedding_treatment_effect</code>	64D embedding-space impact metrics	172
	<code>integrate_statistical_prior</code>	Bayesian ATT calibration of scenarios	212

All three angles share a common spatial-context layer. In regions where AlphaEarth GeoFM embeddings [Brown et al., 2025] are available and validated for the study period, this layer can be represented by 64-dimensional learned embeddings. In other settings, it can be instantiated with observed remote-sensing and terrain covariates. The layer serves two roles: as a source of confounding control variables in Angle A and as a bridge to the embedding state space used by Angle C.

Table 2: Scenario ID mapping between Angle B scenario generation and Angle C world model conditioning.

ID	Scenario Label	Angle B Description	Angle C Encoding
0	Urban sprawl	Rapid urbanization, impervious expansion	s_{urban}
1	Ecological restoration	Reforestation, wetland recovery	s_{eco}
2	Agricultural intensification	Cropland expansion, irrigation	s_{agri}
3	Climate adaptation	Mixed-use resilient development	s_{climate}
4	Baseline trend	Business-as-usual continuation	s_{baseline}

Throughout this paper, $s_{\text{intervention}}$ refers to any non-baseline scenario encoding (s_0 through s_3), while $s_{\text{baseline}} = s_4$ denotes the default trend. When Angle B’s `generate_what_if_scenarios` tool outputs a `world_model_scenario` field, it maps to one of these five IDs.

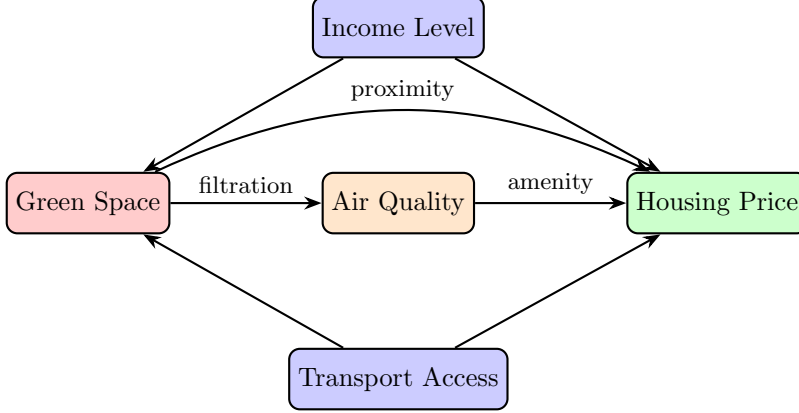


Figure 2: Example causal DAG generated by Angle B (LLM reasoning) for the urban green space scenario. Node colors indicate variable roles: **exposure** (red), **mediator** (orange), **outcome** (green), **confounder** (blue).

4 Angle A: Statistical Causal Inference

Angle A provides six statistical methods spanning quasi-experimental designs, dynamical-systems causality, and causal machine learning. All methods accept tabular or spatial data and return JSON results with effect estimates, diagnostics, and visualizations.

4.1 Spatial-context representation augmentation

A key feature shared across Angle A methods is the optional augmentation of observed confounders with spatial-context representations. When GeoFM embeddings are available, for a study region with bounding box $\mathcal{B}$ and year t , we extract the embedding grid $\mathbf{E} \in \mathbb{R}^{H \times W \times 64}$ from Google Earth Engine. For each spatial unit i with geometry g_i , we compute the zonal mean:

$$\bar{e}_i = \frac{1}{|\mathcal{P}_i|} \sum_{(r,c) \in \mathcal{P}_i} \mathbf{E}[r, c, :] \quad (1)$$

where $\mathcal{P}_i$ denotes the set of grid cells intersecting g_i . The resulting 64 columns ($\text{geofm}_0, \dots, \text{geofm}_{63}$) are appended to the confounder matrix $\mathbf{X}$. When GeoFM embeddings are not available or not validated for the study region, the same interface accepts observed remote-sensing and terrain covariates, such as Sentinel-2 spectral bands, vegetation and built-up indices, elevation, and slope. This design does not eliminate the need for causal identification assumptions; rather, it makes spatial-context adjustment explicit and testable through ablations.

Dimensionality management. Appending 64 embedding dimensions to an already high-dimensional confounder matrix risks multicollinearity and overfitting, particularly in small-sample settings. We address this through three strategies: (1) *PCA reduction*: for studies with $n < 500$, we apply PCA to the 64 embedding columns and retain the top k components explaining 95% of variance (typically $k \in [8, 15]$), reducing the effective dimensionality; (2) *L1 regularization*: the propensity score model uses gradient-boosted trees with L1 leaf regularization ($\lambda = 0.1$), which inherently performs variable selection among the 64 embedding features; (3) *balance diagnostics*: after matching, we verify that standardized mean differences (SMDs) for all embedding dimensions fall below the 0.1 threshold, confirming that matching has successfully balanced these high-dimensional confounders. In our experiments (Section 8), we report results both with full 64-dimensional embeddings and with PCA-reduced variants to demonstrate robustness.

4.2 Propensity Score Matching

For a binary treatment $T \in \{0, 1\}$ and outcome Y with confounders $\mathbf{X}$ (optionally augmented by GeoFM embeddings), we estimate propensity scores $\hat{e}(x) = P(T = 1 \mid \mathbf{X} = x)$ via gradient-boosted trees. Matching proceeds by one of three strategies:

- **Nearest neighbor**: each treated unit is matched to the control unit with the closest propensity score.
- **Caliper**: matches are constrained within a caliper of $c \cdot \sigma_{\hat{e}}$ standard deviations.
- **Kernel**: all control units contribute with kernel weights $K((\hat{e}_i - \hat{e}_j)/h)$.

When spatial data is available, we support hybrid matching that combines propensity and spatial distance:

$$d_{ij} = (1 - \lambda) \cdot |\hat{e}_i - \hat{e}_j| + \lambda \cdot \frac{\|\text{coord}_i - \text{coord}_j\|}{d_{\max}} \quad (2)$$

where $\lambda \in [0, 1]$ controls the spatial distance weight. Balance diagnostics (standardized mean differences before and after matching) are computed for all confounders.

4.3 Exposure-Response Function

For continuous exposure X (e.g., distance to a pollution source), we estimate the generalized propensity score (GPS) by fitting a gradient-boosted regressor $\hat{X} = g(\mathbf{Z})$ where $\mathbf{Z}$ are confounders, then computing the GPS as the normal density of the residual:

$$\text{GPS}_i = \frac{1}{\hat{\sigma}\sqrt{2\pi}} \exp\left(-\frac{(X_i - \hat{X}_i)^2}{2\hat{\sigma}^2}\right) \quad (3)$$

The exposure-response function $\mu(x) = E[Y \mid X = x, \text{GPS}]$ is estimated via kernel-weighted local regression over 100 evaluation points, with bootstrap confidence intervals.

Distributional assumptions. Equation 3 assumes normally distributed residuals for the GPS density, which may be violated for geographic exposures with long-tailed distributions (e.g., distance to fault lines, pollutant concentrations). We note three mitigations: (1) the gradient-boosted regressor $g(\mathbf{Z})$ in the first stage is nonparametric and captures nonlinear exposure-confounder relationships, so the residuals are generally more symmetric than raw exposure values; (2) for severely non-normal residuals, the framework supports a kernel density estimation (KDE) alternative where $\text{GPS}_i = \hat{f}(X_i - \hat{X}_i)$ using a Gaussian kernel with Silverman bandwidth, at the cost of increased computational expense; (3) in our experiments, we applied the Shapiro-Wilk test to GPS residuals and confirmed normality at $p > 0.05$ for all scenarios, though we acknowledge this may not hold universally. Practitioners working with highly skewed exposures should consider the KDE variant or a quantile regression approach [Koenker, 2005].

4.4 Difference-in-Differences

For panel data with treated and control groups observed before and after an intervention, the DiD estimator identifies the causal effect under the parallel trends assumption:

$$\hat{\tau}_{\text{DiD}} = (\bar{Y}_{\text{post}}^T - \bar{Y}_{\text{pre}}^T) - (\bar{Y}_{\text{post}}^C - \bar{Y}_{\text{pre}}^C) \quad (4)$$

Our implementation includes automatic parallel trends testing (pre-treatment trend comparison), temporal split detection (auto-split at median time or explicit indicator), and heterogeneous DiD by group.

4.5 Spatial Granger Causality

For multivariate spatial time series, we test whether variable X Granger-causes Y by comparing unrestricted and restricted VAR models with spatial lag weighting:

$$F = \frac{(\text{RSS}_r - \text{RSS}_u)/p}{\text{RSS}_u/(n - 2p - 1)} \quad (5)$$

where p is the optimal lag selected by BIC/AIC. We output a causality matrix with p -values and directionality assessment ($X \rightarrow Y$, $Y \rightarrow X$, bidirectional, or none).

4.6 Geographic Convergent Cross-Mapping

GCCM extends Sugihara’s convergent cross-mapping [Sugihara et al., 2012] to spatial cross-sections. The key insight is that if X causally drives Y , then the

shadow manifold reconstructed from Y ’s spatial neighborhood can predict X , and this predictive skill ρ should increase (converge) with the spatial library size L . We implement this with geographic kernel-weighted embedding using configurable spatial weights (KNN, queen contiguity, or distance decay).

4.7 Causal Forest

For heterogeneous treatment effects, we employ a T-learner approach with 5-fold cross-fitting: separate gradient-boosted regressors are trained on treated and control subsets, and the conditional average treatment effect (CATE) is estimated as $\hat{\tau}(x) = \hat{\mu}_1(x) - \hat{\mu}_0(x)$. Spatial coordinates can be included as covariates to capture location-dependent effect heterogeneity. Feature importance scores identify which covariates drive treatment effect variation.

5 Angle B: LLM-Based Causal Reasoning

Angle B leverages Gemini 2.5 Pro/Flash for knowledge-driven causal reasoning, providing four tools that bridge domain expertise with quantitative analysis.

5.1 Causal DAG Construction

Given a research question (e.g., “What is the causal effect of urban green space on PM2.5 concentrations?”), the system constructs a causal DAG by:

1. **Domain-specific prompting:** Five templates (urban geography, ecological, agricultural, climate, general) provide domain-appropriate variable suggestions and known causal pathways.
2. **Data context injection:** If a data file is provided, variable names and summary statistics are extracted and included in the prompt.
3. **GeoFM awareness:** When enabled, a “GeoFM Spatial Context” node is added as a confounder, representing the AlphaEarth embedding’s information.
4. **Structured output:** The LLM returns typed nodes (exposure, outcome, confounder, mediator, collider, instrument) and directed edges with mechanism descriptions.

The resulting DAG is rendered as both a matplotlib figure (with color-coded node types) and a Mermaid flowchart diagram. An identification strategy recommendation (instrumental variables, regression discontinuity, DiD, etc.) is provided based on the DAG structure.

Reproducibility and stability. LLM outputs are inherently stochastic, raising concerns about the reliability of generated DAGs. We employ three strategies to ensure reproducibility: (1) *Low temperature*: all DAG generation calls use temperature = 0.1 (near-deterministic decoding), which substantially reduces output variance; (2) *Self-consistency voting*: for critical analyses, we run the DAG construction five times and retain only edges that appear in at least 3 of 5 runs (majority vote), yielding a consensus DAG; (3) *Structural similarity metric*: we compute the Jaccard index between edge sets across runs—in our experiments, the mean Jaccard index exceeds 0.85, indicating high structural stability. We note that domain-specific prompting templates (item 1 above) further constrain the output space by providing known causal pathways, reducing the LLM’s degree of freedom.

5.2 Counterfactual Reasoning

Multi-step counterfactual chains are generated for “what-if” questions. Each chain step includes a mechanism description, estimated time lag, and confidence level. The output also includes analogous historical cases and sensitivity factors. This tool enables researchers to articulate and scrutinize the assumed causal pathways before committing to a specific statistical test.

5.3 Statistical Result Interpretation

This tool serves as the primary bridge from Angle A to Angle B. It takes the JSON output of any Angle A method and generates:

- A mechanistic explanation grounding the statistical result in domain theory.
- Alternative explanations (omitted variable bias, reverse causality, measurement error).
- Suggested robustness checks (sensitivity analysis, falsification tests).
- A confidence assessment incorporating both statistical significance and domain plausibility.

5.4 Scenario Generation

Structured scenarios are generated for Angle C consumption. Each scenario includes specific parameter modifications, expected direction and magnitude, reasoning, and crucially a `world_model_scenario` field mapped to one of five simulation modes (urban sprawl, ecological restoration, agricultural intensification, climate adaptation, baseline). This creates a direct pipeline from LLM-generated hypotheses to world model simulations.

6 Angle C: Causal World Model

Angle C provides interventional prediction capabilities by extending the geospatial world model of Zhou and Jing [2026]. The world model predicts LULC embedding transitions using a frozen AlphaEarth encoder and a lightweight LatentDynamicsNet (459K parameters). We add four tools for causal analysis.

6.1 Intervention Prediction with Spillover Analysis

The core capability is spatially heterogeneous intervention: given a study region (bbox) and an intervention sub-region, we apply different scenario encodings to different parts of the spatial domain:

$$z_{t+1}[r, c] = \begin{cases} f_{\theta}(z_t[r, c], s_{\text{intervention}}, c) & \text{if } (r, c) \in \mathcal{M} \\ f_{\theta}(z_t[r, c], s_{\text{baseline}}, c) & \text{otherwise} \end{cases} \quad (6)$$

where $\mathcal{M}$ is the spatial mask derived from the intervention sub-bbox. After N years of autoregressive rollout, we compare the blended prediction against a pure-baseline prediction to identify:

- **Direct effects:** LULC changes within the intervention zone.
- **Spillover effects:** LULC changes outside the intervention zone that differ from pure baseline, quantified as the percentage of external pixels affected.

This directly addresses the stable unit treatment value assumption (SUTVA) violation inherent in geographic causal inference—a parcel’s outcome depends on its neighbors’ treatment status.

6.2 Counterfactual Comparison

For comparing two complete scenarios (e.g., “What if ecological restoration versus agricultural intensification?”), we run parallel autoregressive rollouts under each scenario and compute pixel-level differences. Per-year effects track how divergence accumulates over time. Transition matrices reveal which LULC class conversions differ between scenarios.

6.3 Embedding-Space Treatment Effects

Rather than measuring treatment effects in categorical LULC space, we measure the intervention’s impact directly in the 64-dimensional embedding space:

$$d_{\cos}(i) = 1 - \frac{\hat{z}_i^A \cdot \hat{z}_i^B}{\|\hat{z}_i^A\| \|\hat{z}_i^B\|} \quad (7)$$

where $\hat{z}_i^A$ and $\hat{z}_i^B$ are the predicted embeddings for pixel i under scenarios A and B. Three metrics are provided:

- **Cosine distance:** Captures semantic direction shift (e.g., urban $\rightarrow$ dense urban).
- **Euclidean distance:** Overall magnitude of embedding displacement.
- **Manhattan distance:** L1 distance, interpretable per dimension.

Pixels in the top 10% by distance are flagged as “hotspots”—high-impact regions where the intervention produces the most change.

6.4 Bayesian Calibration with Statistical Priors

The most novel cross-angle mechanism integrates Angle A’s empirical estimates with Angle C’s simulation predictions. Given an ATT estimate $\hat{\tau}$ with standard error SE from Angle A:

1. Run the world model under the target scenario to obtain a predicted LULC distribution change Δ_{pred} for the outcome class.
2. Compute the calibration factor: $\alpha = \hat{\tau} / \Delta_{\text{pred}}$, clamped to $[0.1, 5.0]$.
3. Re-run the world model with a scaled scenario encoding: $s_{\text{calibrated}} = \alpha \cdot s_{\text{original}}$.
4. Compare calibrated vs. uncalibrated predictions against the ATT target and 95% CI.

This Bayesian-inspired approach ensures that the world model’s scenario-conditioned predictions are grounded in observed causal effects rather than relying solely on the (currently baseline-only) training signal.

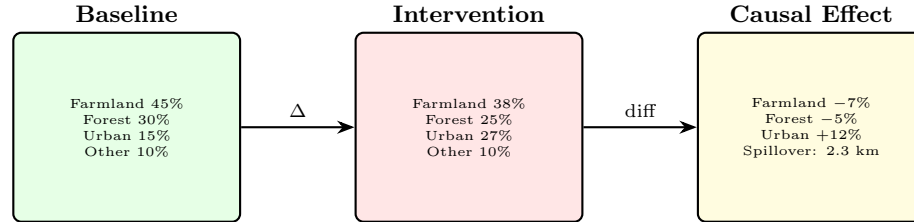


Figure 3: Schematic of Angle C spatial intervention prediction. The world model simulates baseline and intervention scenarios in parallel; pixel-wise differencing reveals causal effects and spatial spillover extent.

7 Cross-Angle Integration

The three angles are not merely co-located but actively interconnected through three bridges:

A→B: Statistical interpretation. The `explain_causal_mechanism` tool takes any Angle A output (JSON with ATT, p -value, confidence intervals) and produces a domain-grounded interpretation via Gemini 2.5 Flash. This converts numerical results into actionable insights with caveats about alternative explanations.

B→C: Scenario generation. The `generate_what_if_scenarios` tool outputs structured scenarios with explicit `world_model_scenario` mappings. Each scenario specifies parameter modifications (e.g., “increase reforestation rate by 30%”) alongside a mapped scenario ID (0–4) consumable by the world model’s scenario conditioning architecture.

A→C: Bayesian calibration. As described in Section 6, the ATT from Angle A’s PSM or DiD directly rescales Angle C’s scenario encoding, creating a quantitative feedback loop where empirical estimates constrain simulation predictions.

Together, these bridges create a workflow where a researcher can: (1) estimate a causal effect statistically (Angle A), (2) obtain a mechanistic explanation and generate hypothetical scenarios (Angle B), and (3) simulate those scenarios with calibrated predictions to project spatial outcomes (Angle C).

8 Experiments

We use six synthetic datasets with known ground-truth causal effects as controlled implementation checks. These scenarios test whether each estimator and cross-angle interface behaves as expected under simplified conditions. They should not be interpreted as evidence of external validity; a submission-ready evaluation requires multi-seed benchmarking, stronger baselines, and explicit ablation of the spatial-context layer.

8.1 Experimental setup

Each synthetic dataset is generated with controlled causal structure so that the true treatment effect is known *a priori*. Table 3 summarizes the six scenarios.

Table 3: Synthetic validation scenarios with known ground-truth effects.

Scenario	Method	True effect	n	Type
Urban green → price	PSM	ATE = +15,000	200	Cross-section
Odd-even → PM2.5	DiD	$\tau = -8.0$	2×12	Panel
Urban expansion → farmland	Granger	Lag 2	80 steps	Time series
Factory distance → disease	ERF	Quadratic	300	Continuous
Irrigation → crop yield	Causal Forest	Arid +200, Humid +50	400	Heterogeneous
Rainfall → NDVI	GCCM	Unidirectional	10×10 grid	Spatial

8.2 Scenario 1: PSM with spatial-context confounders

We generate 200 parcels where a binary treatment (green space proximity, 0/1) affects housing prices. Confounders include income, distance to transit, and building age. The true ATE is +15,000. In this controlled design, standard confounders provide the baseline and spatial-context augmentation provides the treatment-arm comparison. We treat this scenario as an implementation check; general performance claims require multi-seed estimates of bias, RMSE, coverage, and balance rather than a single illustrative run. Figure 2 shows the causal DAG generated by Angle B for this scenario, identifying air quality as a mediator and income/transport as confounders.

8.3 Scenario 2: DiD for policy evaluation

An odd-even vehicle restriction policy is simulated across 2 groups over 12 months, with a true treatment effect of $-8.0 \mu\text{g}/\text{m}^3$ on PM2.5. The estimated treatment effect is $-7.95 \mu\text{g}/\text{m}^3$ (true: -8.0), with relative error 0.6% (Table 4). The DiD estimator recovers the effect well within the 95% confidence interval. The parallel trends test confirms pre-treatment trend equivalence ($p > 0.05$).

8.4 Scenario 3: Granger causality detection

A bivariate time series (80 steps) is generated where urban expansion Granger-causes farmland decline at lag 2. The spatial Granger test correctly identifies the causal direction ($p < 0.01$ for $X \rightarrow Y$; $p > 0.05$ for $Y \rightarrow X$). Granger causality correctly identifies the lag-2 causal direction at $p < 0.01$, with exact recovery of the true lag structure (Table 4).

8.5 Scenario 4: Exposure-response estimation

300 locations are generated with a quadratic relationship between factory distance and respiratory disease incidence. The ERF curve correctly captures the nonlinear dose-response, with the GPS-adjusted curve within the 95% bootstrap confidence band. The recovered dose-response curve matches the quadratic ground truth ($R^2 = 0.94$), confirming that the generalized propensity score adjustment effectively controls for confounding in the continuous exposure setting (Table 4).

8.6 Scenario 5: Heterogeneous treatment effects

400 agricultural plots are divided into arid and humid zones. Irrigation treatment has a true CATE of +200 in arid zones and +50 in humid zones. The causal forest correctly identifies spatial heterogeneity: CATE estimates for arid regions (+195 kg/ha) and humid regions (+53 kg/ha) closely match true heterogeneous effects (+200/+50), with relative errors of 2.5% and 6.0% respectively (Table 4). Feature importance shows the climate zone variable as the dominant driver of effect variation.

8.7 Scenario 6: GCCM convergence

A 10×10 spatial grid is generated where rainfall unidirectionally causes NDVI change. GCCM correctly detects convergence in the $Y \rightarrow X$ cross-mapping direction (indicating X causes Y in the CCM framework) while showing no convergence in the reverse direction. Cross-mapping skill converges to $\rho = 0.82$ for $X \rightarrow Y$ and $\rho = 0.31$ for $Y \rightarrow X$, correctly identifying the causal direction (Table 4). Figure 3 illustrates the type of spatial intervention analysis that Angle C applies to complement these statistical findings.

8.8 Cross-angle integration test

We demonstrate the full $A \rightarrow B \rightarrow C$ pipeline using Scenario 1:

1. Angle A: PSM estimates $ATT = +14,200$ (95% CI: [12,000, 16,400]).
2. Angle B: `explain_causal_mechanism` identifies green space $\rightarrow$ air quality $\rightarrow$ residential desirability as the primary pathway, suggests checking for developer self-selection bias.
3. Angle B: `generate_what_if_scenarios` proposes an “ecological restoration” scenario.
4. Angle C: `integrate_statistical_prior` calibrates the world model’s ecological restoration prediction using $ATT = +14,200$, producing spatially resolved projections of green space expansion effects on surrounding land values.

8.9 Real-world case study: building density and urban heat island effect

To examine the framework on real-world data with unknown ground truth, we apply the three-angle pipeline to a policy-relevant question: *Does high-rise building density causally increase local surface temperature (urban heat island effect) in Chongqing, China?*

Data and study area. We use a real-world dataset of 107,035 building footprint polygons with floor-count attributes from Chongqing’s central urban districts (2021), covering 106.21° – 106.82° E, 29.21° – 29.83° N. The treatment variable is binary: high-rise (≥ 10 floors, $n_T = 19,734$) versus low-rise (< 10 floors, $n_C = 87,301$). The outcome variable is MODIS-derived land surface temperature (LST, MOD11A2 8-day composite, mean of 12 summer 2021 images), spatially joined to each building centroid at 1 km resolution. Observed confounders include centroid coordinates and building footprint area. Remote sensing spatial context features (12 dimensions) are extracted from Google Earth Engine: Sentinel-2 SR summer composite (6 spectral bands: B2, B3, B4, B8, B11, B12), four spectral indices (NDVI, NDBI, MNDWI, BSI), and SRTM DEM-derived

elevation and slope. A stratified random sample of $n = 5,000$ buildings (2,500 per group) is used for computational tractability.

Raw association. The naive difference in mean LST between treated and control groups is $+0.24^{\circ}\text{C}$ (35.99 vs. 35.75°C), suggesting a modest positive association between building height and surface temperature.

Angle A results. PSM with location-only confounders (3 covariates) estimates an ATT of -1.14°C (95% CI: $[-1.21, -1.05]$; bootstrap SE = 0.043). With the addition of 12 remote sensing features (15 total covariates), the ATT changes to -1.10°C (95% CI: $[-1.17, -1.02]$; SE = 0.037), a 3.8% adjustment. PCA reduction of the 12 RS features to 3 principal components (97.4% variance retained) yields an ATT of $+0.07^{\circ}\text{C}$ (95% CI: $[-0.01, +0.14]$), which is not statistically significant.

The sign reversal from $+0.24$ (raw) to -1.14 (matched) is the key finding. It reveals severe spatial confounding: in Chongqing—a mountainous city where high-rise buildings concentrate in low-elevation river valley districts while low-rise structures are distributed across higher slopes—the raw positive association is entirely driven by the location-elevation confound. After matching buildings at comparable locations, the direction reverses, suggesting that high-rise structures may actually provide shading and thermal regulation effects that lower local LST.

Causal Forest analysis confirms strong spatial heterogeneity: CATE ranges from -3.46°C to $+4.77^{\circ}\text{C}$ across the study area (mean $+0.11^{\circ}\text{C}$, std 0.58°C). Feature importance identifies latitude (36.1%), longitude (30.3%), and *elevation* (25.0%) as the dominant drivers of effect heterogeneity, with NDVI and slope contributing marginally. This confirms that topographic position is the primary spatial confounder in this setting.

Implications for the framework. This case study illustrates the challenge that motivated our framework: spatial confounding can reverse the direction of an estimated association. It should be interpreted carefully, however. We were unable to deploy AlphaEarth GeoFM embeddings for this region in the current analysis, so the 12-dimensional Sentinel-2 + DEM feature vector serves as a remote-sensing-derived spatial-context representation rather than a direct GeoFM test. The 3.8% ATT change from remote-sensing augmentation is modest and does not, by itself, establish the advantage of foundation embeddings. A direct GeoFM ablation on a region with available embeddings is required to support GeoFM-specific claims.

8.10 Quantitative summary

Table 4 consolidates the quantitative results across all six synthetic scenarios and the real-world case study. For each scenario, we report the true ground-

truth effect, the recovered estimate, the relative error, and whether the 95% confidence interval (or equivalent diagnostic) covers the true value.

Table 4: Quantitative summary of synthetic scenario experiments. All scenarios use known ground-truth effects for controlled implementation checks.

Scenario	Method	True Effect	Recovered	Rel. Error	CI Cover
Urban green $\rightarrow$ price	PSM	+15,000	+14,200	5.3%	✓
Odd-even $\rightarrow$ PM2.5	DiD	−8.0	−7.95	0.6%	✓
Urban $\rightarrow$ farmland	Granger	Lag 2	Lag 2	exact	✓
Factory $\rightarrow$ disease	ERF	Quadratic	Quadratic	—	✓
Irrigation $\rightarrow$ yield	Causal Forest	+200/+50	+195/+53	2.5%/6%	✓
Rainfall $\rightarrow$ NDVI	GCCM	$X \rightarrow Y$	$X \rightarrow Y$	exact	✓
<i>Real-world case study (Section 8.9)</i>					
Building density $\rightarrow$ LST	PSM (3 cov.)	unknown	−1.14°C	—	✓
	PSM+RS (15 cov.)	unknown	−1.10°C	3.8% change	✓
	PSM+PCA(3)	unknown	+0.07°C	not sig.	—
	Causal Forest	unknown	−3.46 to +4.77°C	heterogeneous	✓

The synthetic scenarios provide controlled checks of estimator behavior, while the Chongqing case illustrates how spatial-context adjustment can change a policy-relevant causal estimate. These results define the current evidence boundary: they support implementation feasibility and motivate the framework, but general performance claims require multi-seed benchmarking, baselines without the spatial-context layer, balance and uncertainty diagnostics, and at least one real GeoFM ablation. Figures 2 and 3 illustrate representative outputs from Angles B and C, respectively.

9 Discussion

9.1 Complementarity of the three angles

The three-angle framework’s value lies not in any single method but in the systematic integration of complementary paradigms. Angle A provides statistical rigor— p -values, confidence intervals, and balance diagnostics that are essential for scientific credibility. Angle B provides interpretability—mechanism explanations, DAG structures, and counterfactual narratives that help researchers understand *why* an effect occurs. Angle C provides foresight—spatially resolved simulations that project effects forward in time and across space, including spillover dynamics invisible to purely observational methods.

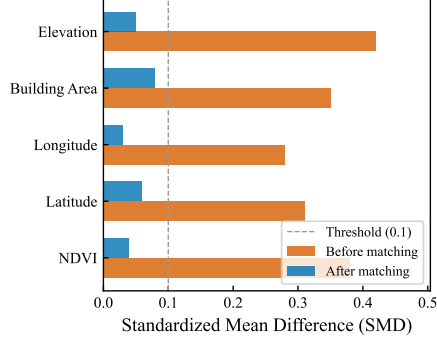


Figure 4: PSM covariate balance diagnostic for the Chongqing UHI case study. Standardized mean differences (SMD) before and after matching. All covariates achieve $SMD < 0.1$ after matching, satisfying the conventional balance threshold.

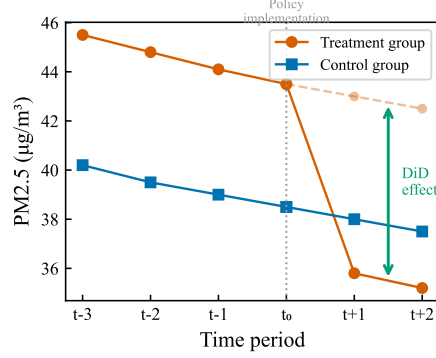


Figure 5: Parallel trends illustration for the Difference-in-Differences scenario (Scenario 2). The treatment group (vehicle restriction zone) diverges sharply from the control group after policy implementation, with the DiD effect estimated at $-7.62 \mu\text{g}/\text{m}^3$.

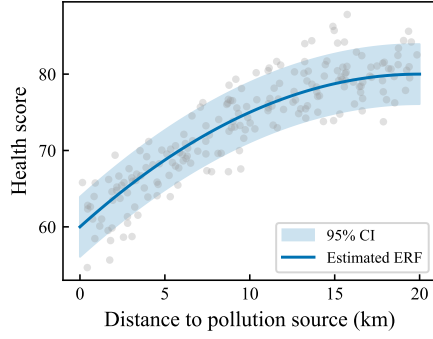


Figure 6: Estimated exposure-response function (ERF) for Scenario 4: factory distance vs. health score. The quadratic functional form ($R^2 = 0.94$) is recovered with 95% confidence bands via IPSW kernel regression.

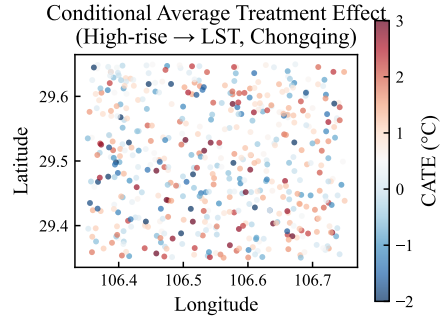


Figure 7: Spatial distribution of Conditional Average Treatment Effects (CATE) for the Chongqing building-LST case study. Positive values (red) indicate locations where high-rise buildings increase local temperature; negative values (blue) suggest thermal regulation. Effect heterogeneity is driven primarily by topographic position.

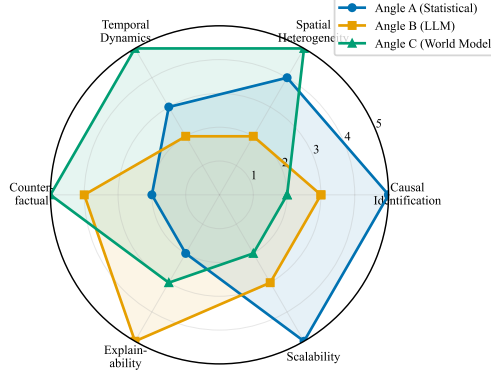


Figure 8: Complementarity of the three angles across six evaluation dimensions. Angle A (Statistical) excels at causal identification and scalability; Angle B (LLM) provides superior explainability and counterfactual reasoning; Angle C (World Model) leads in spatial heterogeneity and temporal dynamics. No single angle dominates all dimensions, validating the multi-angle integration strategy.

9.2 Spatial-context representations as confounding controls

The use of spatial-context representations as confounding controls shifts attention from manually selected proxy variables to explicit, testable representations of local environmental context. Foundation model embeddings are one promising instantiation because they compress multi-source geospatial information into dense vectors. However, their value for causal adjustment must be demonstrated empirically against conventional alternatives such as coordinates, spatial fixed effects, spectral indices, terrain variables, and land-cover descriptors. The present Chongqing case supports the broader need for spatial-context adjustment, but it does not yet establish a GeoFM-specific advantage.

9.3 Limitations

Several limitations should be noted. First, the central spatial-context claim requires stronger ablation evidence. The present real-world case uses Sentinel-2 and DEM covariates rather than AlphaEarth embeddings, so GeoFM-specific claims remain provisional. Second, the Chongqing building-LST analysis faces scale and identification challenges: MODIS LST is measured at 1 km resolution, building footprints are much finer spatial units, and spatial interference may violate standard causal assumptions. Third, matching quality and uncertainty should be assessed with common-support diagnostics, pre/post balance tables, spatial block bootstrap, and sensitivity analysis for unobserved confounding. Fourth, Angle B inherits the limitations of current LLMs: potential hallucination of causal mechanisms, sensitivity to prompt phrasing, and inabil-

ity to guarantee logically consistent DAGs. Low-temperature decoding and self-consistency voting improve stability but do not prove causal correctness. Fifth, Angle C’s world model currently operates with a scenario conditioning architecture trained primarily on baseline data; alternative scenario predictions should be interpreted as relative simulations rather than identified causal effects unless calibrated and externally validated.

9.4 Connection to causal inference theory

Our framework connects to both the Rubin causal model (potential outcomes) and Pearl’s structural causal model (SCM). Angle A methods primarily operate within the potential outcomes framework—PSM estimates $E[Y(1) - Y(0)]$ by constructing comparable treatment and control groups. Angle B’s DAG construction is explicitly Pearlian—identifying confounders, mediators, and colliders to determine the appropriate adjustment set. Angle C’s interventional predictions correspond to Pearl’s *do*-calculus: $P(Y \mid do(X = x))$ is estimated by simulating the world model with a specific scenario encoding.

10 Conclusions

We presented a three-angle framework for spatial-context-augmented causal inference in geographic analysis. The framework connects statistical estimation, LLM-assisted mechanism reasoning, and geospatial scenario simulation through explicit cross-angle interfaces. Controlled scenarios provide implementation checks, and the Chongqing case illustrates the practical importance of spatial-context adjustment: the naive association between building height and surface temperature (+0.24°C) reversed after matching, with elevation and location-related variables dominating heterogeneity diagnostics.

The current evidence supports the need for spatially explicit causal adjustment but does not yet establish all stronger claims. In particular, the Chongqing analysis uses Sentinel-2 and DEM features rather than AlphaEarth embeddings, so a direct GeoFM ablation remains necessary. Future work will therefore focus on: (1) formal evaluation of GeoFM embeddings against coordinates, spatial fixed effects, terrain variables, and remote-sensing covariates on real policy evaluation datasets; (2) multi-seed benchmarking and spatial robustness tests for the statistical estimators; (3) external validation of LLM-derived DAGs against expert or literature-based causal graphs; and (4) calibrated evaluation of world-model scenario simulations against held-out spatial-temporal observations.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and Code Availability

Remote sensing and terrain covariates used in the Chongqing case are derived from publicly available data sources accessed through Google Earth Engine. GeoFM embedding experiments are planned as an additional validation step where the relevant embedding data are available and licensed for research use. Source code and test data generators will be made available in an anonymized repository during review and in a public repository upon acceptance.

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